

Dataset Shape: (150, 6)

First 5 Rows:

	Id	SepallLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

Dataset Info:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 150 entries, 0 to 149

Data columns (total 6 columns):

#	Column	Non-Null Count	Dtype
0	Id	150 non-null	int64
1	SepallLengthCm	150 non-null	float64
2	SepalWidthCm	150 non-null	float64
3	PetalLengthCm	150 non-null	float64
4	PetalWidthCm	150 non-null	float64
5	Species	150 non-null	object

dtypes: float64(4), int64(1), object(1)

memory usage: 7.2+ KB

None

Class Distribution:

Species

Iris-setosa 50

Iris-versicolor 50

Iris-virginica 50

Name: count, dtype: int64

Accuracy for Different K Values

K = 1 --> Accuracy = 0.9667
K = 3 --> Accuracy = 0.9333
K = 5 --> Accuracy = 0.9333
K = 7 --> Accuracy = 0.9667
K = 9 --> Accuracy = 0.9667

Final Accuracy: 0.9333333333333333

Classification Report:

	precision	recall	f1-score	support
Iris-setosa	1.00	1.00	1.00	10
Iris-versicolor	0.83	1.00	0.91	10
Iris-virginica	1.00	0.80	0.89	10
accuracy			0.93	30
macro avg	0.94	0.93	0.93	30
weighted avg	0.94	0.93	0.93	30



